

AlgoAds — Google Ads API Design Document

Company: Algo Thinker **Product:** AlgoAds — Demand Gen Campaign Automation for YouTube Video Promotion
Website: <https://algo-thinker.com> **API Use Case Page:** <https://algo-thinker.com/ads-api> **Privacy Policy:** <https://algo-thinker.com/privacy> **Terms of Service:** <https://algo-thinker.com/terms> **Developer Contact:** api@algo-thinker.com **Date:** April 2026

1. Product Overview

AlgoAds is a web-based tool that automates the creation of Google Ads Demand Gen video campaigns for YouTube content creators. It reduces the campaign setup time from 1-2 hours (manual Google Ads UI) to under 5 minutes while keeping the creator in full control of their account, budget, and creative.

Target users: YouTube content creators who want to promote their own videos via Google Ads but lack the expertise to navigate the full Google Ads interface.

Business model: AlgoAds is currently free during beta. Revenue model TBD (potential SaaS subscription for creators). AlgoAds is NOT a Google Ads reseller and does NOT handle billing.

2. User Flow

Step 1: Authentication

- User signs into AlgoAds via email/password (managed by InsForge auth backend)
- User connects their Google Ads account via OAuth 2.0 (standard Google consent flow)
- AlgoAds receives a refresh token scoped to the user's Google Ads account
- Refresh token is stored encrypted at rest in our database

Step 2: Creative Generation

- User pastes a YouTube video URL
- AlgoAds reads public video metadata (title, description, thumbnail) via YouTube oEmbed
- AlgoAds sends video metadata to Google Gemini API to generate ad copy:
 - 3-5 headlines (max 40 chars each) - 1-5 long headlines (max 90 chars each) - 1-5 descriptions (max 90 chars each) - Business name suggestion
- User reviews and edits all generated copy before proceeding

Step 3: Campaign Configuration

- User sets daily budget (USD)
- User sets campaign duration (days)
- User uploads a square logo image (min 300x300, PNG or JPG)
- Default targeting: 65 countries (Global South tech markets), English language, YouTube in-feed channel only

Step 4: Campaign Launch via Google Ads API

AlgoAds creates the campaign using a 7-step pipeline, all via the Google Ads REST API v23:

StepAPI EndpointWhat It Creates |-----|-----|-----| 1 `campaignBudgets:mutate` Daily budget resource 2 `campaigns:mutate` Demand Gen campaign (PAUSED by default, targetSpend bidding) 3 `adGroups:mutate` Ad group with channel controls (YouTube in-feed only) 4 `adGroupCriteria:mutate` Geo targeting (65 countries) + language targeting (English) 5 `assets:mutate` YouTube video asset + logo image asset 6 (inline)Text assets (headlines, descriptions) are inline in the ad 7 `adGroupAds:mutate` DemandGenVideoResponsiveAd with all assets

Campaign is created PAUSED. User enables it from the AlgoAds dashboard.

Step 5: Dashboard & Management

- User views campaign list with live metrics queried via GAQL:
- `SELECT campaign.id, campaign.name, campaign.status, metrics.impressions, metrics.clicks, metrics.video_views, metrics.cost_micros FROM campaign WHERE campaign.id = {id}`
 - User can pause, enable, or remove campaigns via `campaigns:mutate` (status change)
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3. Google Ads API Endpoints Used

Read Operations

EndpointPurposeTrigger |-----|-----|-----| `customers/{id}/googleAds:search` (GAQL)Query campaign list, status, metricsDashboard page load `customers/{id}/googleAds:search` (GAQL)Query ad group statusCampaign detail page `geoTargetConstants` Look up geo target IDs for new citiesAdmin function (rare)

Write Operations

EndpointPurposeTrigger |-----|-----|-----| `campaignBudgets:mutate` Create daily budgetUser clicks "Launch" `campaigns:mutate` Create campaign (PAUSED) / change statusUser clicks "Launch" or "Pause"/"Enable" `adGroups:mutate` Create ad group with channel controlsUser clicks "Launch" `adGroupCriteria:mutate` Set geo + language targetingUser clicks "Launch" `assets:mutate` Upload YouTube video + logo imageUser clicks "Launch" `adGroupAds:mutate` Create DemandGenVideoResponsiveAdUser clicks "Launch"

What We Do NOT Use

- We do NOT use the Reporting API for bulk data extraction
 - We do NOT use the Remarketing or Customer Match APIs
 - We do NOT use the Bidding Strategy Service (we use inline campaign-level bidding)
 - We do NOT use the Conversion Tracking API
 - We do NOT access or modify account-level settings
 - We do NOT manage campaigns on behalf of other advertisers
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4. Data Flow Diagram

```
User (YouTube Creator)
|
| 1. Sign in (email/password via InsForge)
| 2. Connect Google Ads (OAuth 2.0)
|
v
AlgoAds Server (Next.js on Vercel)
|
| 3. Store encrypted refresh token
| 4. User pastes YouTube URL
| 5. Generate creative via Gemini API
| 6. User reviews + clicks Launch
|
|--- Google Ads API v23 (REST) -----> User's Google Ads Account
|   - Create budget                      - Campaign runs here
|   - Create campaign (PAUSED)          - Billed to user's payment method
|   - Create ad group                   - User owns everything
|   - Set geo/language targeting
|   - Upload video + logo assets
|   - Create video responsive ad
|
|--- Google Gemini API -----> Generate ad copy suggestions
|   - Send: video title, description      (public metadata only)
|   - Receive: headlines, descriptions
|
|--- YouTube oEmbed -----> Read public video metadata
|   - Send: video URL
|   - Receive: title, thumbnail URL
```

5. Authentication Flow

1. User clicks "Connect Google Ads"
2. AlgoAds redirects to Google OAuth consent screen
 - Scope: <https://www.googleapis.com/auth/adwords>
3. User grants consent
4. Google redirects back with authorization code
5. AlgoAds exchanges code for refresh token (server-side)
6. Refresh token stored encrypted in database
7. On each API call: AlgoAds uses refresh token to get short-lived access token
8. Access token used in Authorization header for Google Ads API calls

User can revoke access at any time via <https://myaccount.google.com/permissions>

6. Data Storage

DataWhere	Stored	Encryption	Retention	
				User email +
	name	InsForge (managed DB)	Encrypted at rest	Until account deletion
	Google Ads refresh token	InsForge (managed DB)	Encrypted at rest	Until revocation or deletion
	Google Ads customer ID	InsForge (managed		

DB)Encrypted at restUntil account deletion Campaign IDs + namesInsForge (managed DB)Encrypted at restUntil account deletion Campaign metricsNOT storedN/AQueried live via GAQL Logo imagesUploaded to user's Google Ads accountN/ADeleted from our server after upload Usage logsVercel server logsStandard30 days

7. Rate Limiting & Error Handling

- AlgoAds creates at most 1 campaign per user action (no bulk operations)
 - API calls are sequential within a pipeline (7 steps)
 - If any step fails, the pipeline rolls back by removing the budget resource
 - Standard exponential backoff on transient errors (429, 503)
 - Daily API call volume: < 100 calls/day (small user base, no batch operations)
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8. Compliance

- **Google API Services User Data Policy:** AlgoAds adheres to the Limited Use requirements. Google user data is used only to provide the campaign management service.
 - **Privacy Policy:** <https://algo-thinker.com/privacy>
 - **Terms of Service:** <https://algo-thinker.com/terms>
 - **Data deletion:** Users can request full account deletion via privacy@algo-thinker.com
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9. Screenshots

The AlgoAds interface consists of four main screens:

1. **Launch page** — User pastes YouTube URL, generates creative with Gemini, reviews headlines/descriptions, sets budget, and clicks Launch 2. **Campaign list** — Shows all campaigns with live metrics (impressions, clicks, cost) 3. **Campaign detail** — Individual campaign view with pause/enable controls 4. **Login/Signup** — Standard auth flow with Google OAuth for Ads connection

Screenshots and a working demo are available at <https://algo-thinker.com> (sign in required for authenticated pages; public pages include the home page, /ads-api, /privacy, and /terms).